AI-Based Food Recognition with Nutrition-Aware Recommendations

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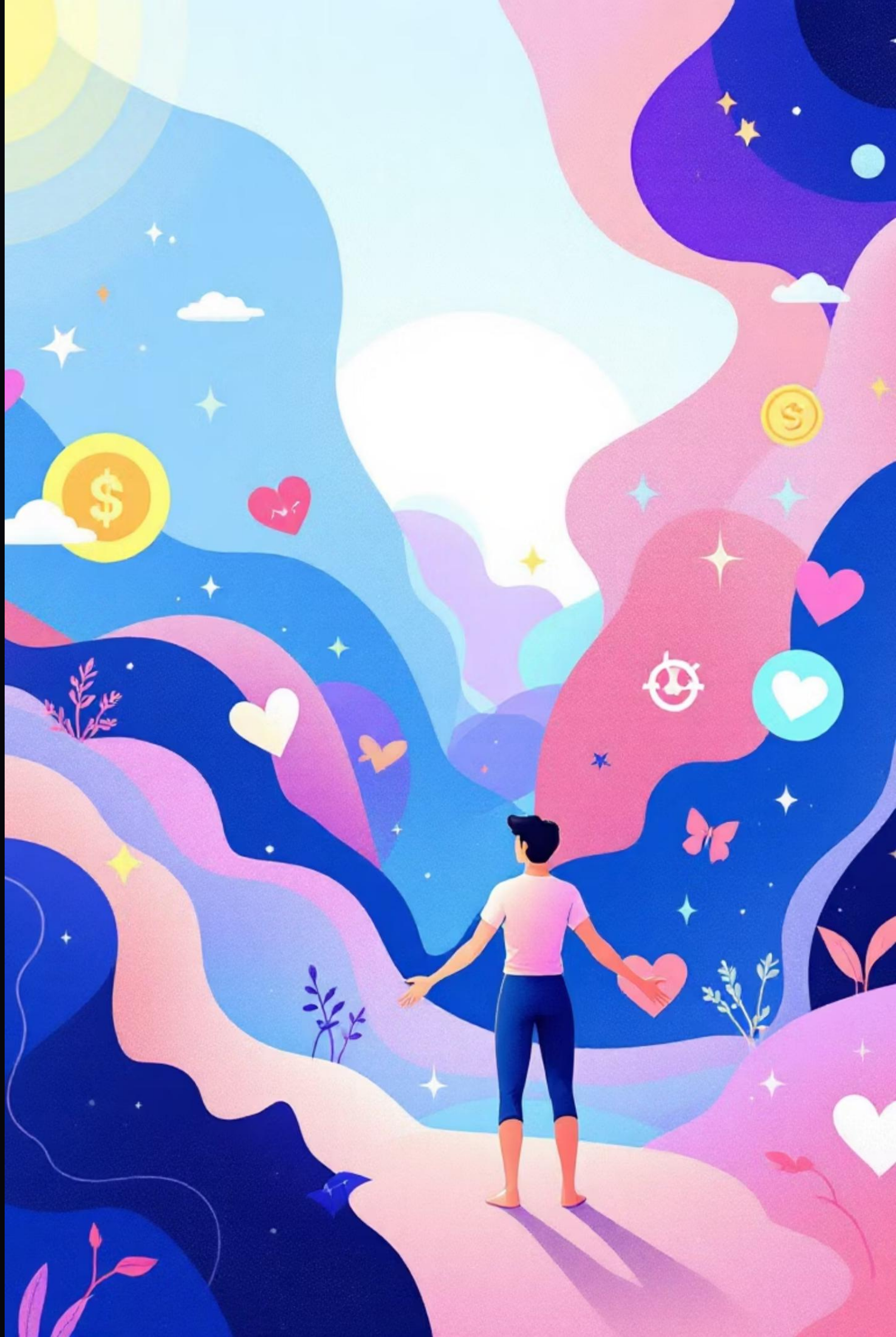
Introduction

For a great life, health, wealth, and relationships matter — with health being the foundation.

Diet monitoring is crucial but often manual and error-prone.

Existing apps are limited: single-food detection, manual calorie logging.

Need: An intelligent system that can detect multiple foods and provide personalized health guidance.



Latest Research in Food Recognition, Nutrition System & Recommendation Systems (Citations)

Agarwal, R., Choudhury, T., Ahuja, N. J., & Sarkar, T. (2023). Hybrid Deep Learning Algorithm-Based Food Recognition and Calorie Estimation. Journal of Food Science and Engineering, 13(6), 1–12

Han, M., Chen, J., & Zhou, Z. (2024). NutrifyAI: An AI-Powered System for Real-Time Food Detection, Nutritional Analysis, and Personalized Meal Recommendations. arXiv preprint arXiv:2408.10532

Papastratis, I., Konstantinidis, D., Daras, P., & Dimitropoulos, K. (2024). AI Nutrition Recommendation Using a Deep Generative Network and ChatGPT. Scientific Reports, 14, 65438

Chougule, S., Patil, V., Deshpande, R., & Rao, M. (2025). Automated Food Recognition and Personalized Health Recommendation System. International Journal of Creative Research Thoughts (IJCRT), 13(2), 45–52

Frontiers in Nutrition. (2025). An AI-Based Nutrition Recommendation System. Frontiers in Nutrition, 12, 1546107

•Existing systems provide robust multi-food detection and general nutrition recommendations.

•They often lack support for regional cuisines and culturally calibrated portion sizes.

•Health-condition–specific guidance (e.g., diabetes, gluten-free, cardiovascular) is minimal.

•Our approach optimizes YOLOv8 for Indian dishes and delivers tailored dietary advice.

Problem Statement

Manual food tracking = **time-consuming and inaccurate**.

Current solutions:

Manual entry of food items

No portion size estimation

No personalized recommendations (BMI, diabetes, gluten-free, dietary preferences)

Gap → AI-based, integrated food detection + nutrition + personalized guidance

Objectives

01 Detect multiple food items in a single photo using YOLOv8.

02 Map detected foods to nutrition datasets (Kaggle, Roboflow, USDA CSV).

03 Provide personalized health guidance & recipe suggestions:

- Adjusted for BMI & health conditions (diabetic, gluten-free, high BP, etc.)
- Consider dietary preferences (veg, vegan, non-veg, keto).
- Suggest healthier alternatives or balanced meal plans

Example output:

> Detected: Rice (Medium), Curry (Small), Bread (1 pc)
>
> Calories: 520 kcal | Protein: 15g | Carbs: 70g | Fat: 20g

Forms the **foundation for personalized health guidance and recommendations**.

System Flow (Architecture)

1. User Input: Food photo + health profile (BMI, dietary preferences, health conditions)
2. Food Recognition (YOLOv8): Detect multiple items with bounding boxes
3. Portion Estimation: Small/Medium/Large per bounding box
4. Nutrition Mapping: Lookup calories & macros from Kaggle/USDA CSV datasets
5. Personalized Analysis: Compare intake with BMI, health condition, dietary preference
6. Recipe / Meal Guidance (LLM): Suggest healthier alternatives, balanced meals, substitutions
7. Output: Meal report + nutrition summary + personalized meal guidance

Datasets & Tech Stack

Tech Stack

Detection: YOLOv8

Backend: FastAPI/Flask

Frontend: Streamlit/React

Recommendation System: LLM + prompt engineering for meal suggestions

Data Storage & APIs: PostgreSQL, Edamam/USDA APIs

Computer Vision Datasets

•Food-101 (Kaggle) – 101 classes, 101 000 images (750 train + 250 test each)

•Indian Food Images (Kaggle) – ~80 classes, 4 000 regional dish images

•Food Classification Dataset (Roboflow Universe) – Bounding-box annotations 5 classes

Nutrition & Portion Data

•USDA FoodData Central (USDA portal) – Calories, macros, micronutrients for 9 000+ items

•Indian Nutrition Dataset (Kaggle) – Calorie, protein, fat, carb values for 200+ Indian dishes

•Nutrition5k (GitHub & Kaggle) – RGB-D images of 5 000 dishes with precise weight and nutrition labels

Contribution

Novel combination of multi-food recognition + portion estimation + nutrition mapping.

Personalized dietary guidance based on BMI, health conditions, and preferences.

Provides practical impact for fitness enthusiasts, patients, and dieticians.

Establishes a foundation for scalable AI-driven dietary assistant.

Expected Outcomes & Future Scope

Expected Outcomes:

- Accurate multi-food detection in a single image
- Portion-aware calorie and macronutrient estimation
- Personalized meal guidance / healthier alternatives
- Prototype web app for interaction

Future Scope (scalable version):

- Weekly meal planner with calorie/health tracking
- ML-based advanced recommendation system (user preferences + history)
- Ingredient substitutions and regional cuisine support
- Integration with fitness apps or wearable health trackers

Conclusion

Bridges AI, computer vision, and nutrition science.

Solves manual food tracking gap by detecting multiple foods with portion-aware nutrition mapping.

Personalized dietary guidance makes it practical and impactful.

Core feature ensures research novelty + real-world usability.

Scalable into a full-stack diet and fitness assistant for future development.